# The University of Faisalabad

Department of Computer Science

BS (AI) 6th Semester (6A)

Data Mining


# Project Submition Report

## Project Title: Adaptive Classification Models for Early Heart Disease Prediction Using UCI Dataset


### Supervisor: Sir Arham



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| # | Name | Registration No. | Email |
|---|---|---|---|
| 1 | Muhammad Hassan | 2023-BS-AI-176 | 2023-bs-ai-176@tuf.edu.pk |
| 2 | Ahtisham Akram | 2023-BS-AI-166 | 2023-bs-ai-166@tuf.edu.pk |
| 3 | Rihab Ahmad | 2023-BS-AI-177 | 2023-bs-ai-177@tuf.edu.pk |
| 4 | Abdullah Mehfooz | 2023-BS-AI-170 | 2023-bs-ai-170@tuf.edu.pk |


## Abstract


This project presents an end-to-end machine learning pipeline for predicting heart disease using the UCI Cleveland Heart Disease dataset (303 patients, 13 clinical features). Eleven classifiers spanning classical and ensemble families were implemented and systematically optimised through exhaustive GridSearchCV hyper-parameter tuning. The strongest individual learners were combined into soft-voting and stacking ensembles to further improve predictive reliability.

All models were evaluated on a held-out test set and validated through 10-fold stratified cross-validation, supplemented by a stability analysis across 30 random splits to ensure honest reporting. The best tuned models achieve 90–93% held-out test accuracy and ROC-AUC scores of up to 0.98, consistent with rigorous published benchmarks on this dataset. The pipeline is fully reproducible and serves as a methodologically sound academic demonstration of professional-grade data mining practice.


## 1. Introduction

Cardiovascular disease remains the leading cause of death worldwide, accounting for an estimated 17.9 million fatalities annually (WHO). Early and accurate identification of at-risk individuals from routine clinical measurements can materially improve patient outcomes and reduce the burden on healthcare systems. Machine learning offers a powerful, data-driven avenue to supplement clinical judgment in this domain.

This project leverages the well-established UCI Cleveland Heart Disease dataset to build, tune, and compare eleven classification models — from classical statistical approaches to modern boosted ensemble methods. Rather than selecting a single algorithm, the study adopts a rigorous comparative methodology with exhaustive hyper-parameter search, cross-validated evaluation, and an explicit robustness analysis across multiple random splits.

### 1.1 Objectives

- Load and rigorously clean the UCI Cleveland Heart Disease dataset, handling hidden missing values and class imbalance.
- Perform thorough exploratory data analysis to understand feature distributions, correlations, and clinical relevance.
- Train eleven classifiers spanning classical (Logistic Regression, KNN, Naive Bayes, Decision Tree, SVM) and advanced/ensemble (Random Forest, Gradient Boosting, AdaBoost, Extra Trees, XGBoost, MLP) families.
- Optimise every model with GridSearchCV hyper-parameter tuning using 5-fold cross-validation on the training partition.
- Construct Soft Voting and Stacking ensembles from the strongest base learners and compare all models across accuracy, precision, recall, F1, and ROC-AUC metrics.

## 2. Background & Related Work

### 2.1 Clinical Context & Dataset

The Cleveland Heart Disease dataset, contributed by the Cleveland Clinic Foundation to the UCI Machine Learning Repository, is the most widely cited benchmark for computational cardiac risk modelling. It contains 303 patient records described by 13 clinical attributes including demographic, symptom-based, and laboratory measurements. The binary target encodes the presence (1) or absence (0) of angiographically confirmed heart disease.

| Feature | Clinical Meaning | Type | Range |
|---|---|---|---|
| age | Patient age in years | Numeric | 29–77 |
| sex | 1 = male, 0 = female | Binary | 0, 1 |
| cp | Chest-pain type (0–3) | Categorical | 0–3 |
| trestbps | Resting blood pressure (mm Hg) | Numeric | 94–200 |
| chol | Serum cholesterol (mg/dl) | Numeric | 126–564 |
| fbs | Fasting blood sugar > 120 mg/dl | Binary | 0, 1 |
| restecg | Resting ECG result | Categorical | 0–2 |
| thalach | Maximum heart rate achieved | Numeric | 71–202 |
| exang | Exercise-induced angina | Binary | 0, 1 |
| oldpeak | ST depression (exercise) | Numeric | 0–6.2 |
| ca | Major vessels (fluoroscopy) | Numeric | 0–3 |
| thal | Thalassemia classification | Categorical | 1–3 |

### 2.2 Related Literature

Detrano et al. (1989), the original contributors, achieved 74–77% accuracy using logistic-type discriminant analysis. Modern studies report higher accuracy via machine learning: Chandrasekhar et al. (2023) achieved 90.16% with logistic regression and 93.44% with a soft-voting ensemble. Random Forest implementations converge near 90%, while SVM reaches 92%. Studies reporting 95–100% typically use a duplicate-inflated 1,025-row dataset where records leak between train and test partitions. This project deliberately uses the clean 303-row Cleveland set.

## 3. Methodology

### 3.1 Pipeline Architecture

The pipeline follows six sequential phases: (1) Environment setup with fixed random seed = 4; (2) Data acquisition from a public mirror; (3) Exploratory data analysis; (4) Preprocessing duplicate removal, hidden-missing imputation, stratified 80/20 train/test split, and StandardScaler fit exclusively on training data; (5) Modelling — baseline training, GridSearchCV tuning, and ensemble construction; (6) Evaluation multi-metric comparison, stability analysis, and literature benchmarking.

### 3.2 Model Portfolio

| Classical Models | Advanced / Ensemble Models | Meta-Ensembles |
|---|---|---|
| Logistic Regression | Random Forest | Soft Voting Ensemble |
| K-Nearest Neighbors | Gradient Boosting | Stacking Ensemble |
| Naive Bayes | AdaBoost | - |
| Decision Tree | Extra Trees | - |
| SVM (RBF) | XGBoost / MLP (Neural Network) | - |

### 3.3 Hyper-parameter Tuning & Ensemble Construction

Every model underwent exhaustive GridSearchCV with 5-fold stratified cross-validation on the training set. The two meta-ensembles were built from the five strongest tuned base learners (Logistic Regression, SVM, Random Forest, Gradient Boosting, KNN). The Soft Voting Ensemble averages predicted class probabilities across all five, while the Stacking Ensemble trains a Logistic Regression meta-learner on out-of-fold predictions, allowing it to learn optimal combination weights.

## 4. Exploratory Data Analysis

After duplicate removal, the dataset contains 302 unique patient records. The class distribution is near-balanced: 165 patients (54.6%) have confirmed heart disease and 137 (45.4%) do not, minimising the need for class-rebalancing techniques.

### 4.1 Key Distributional Findings

- Age: Heart disease patients cluster in the 50–65 range, though overlap with healthy patients is considerable — age alone is not sufficiently discriminative.
- Maximum heart rate (thalach): Patients with heart disease show notably lower maximum heart rate (median ~20 bpm below disease-free patients) — one of the most visually separating features.
- Chest pain type (cp): Asymptomatic chest pain (type 0) appears more frequently in disease-positive patients — a counter-intuitive but clinically documented pattern.
- ST depression (oldpeak): Higher values are strongly associated with disease presence.

### 4.2 Feature Correlation with Target

| Feature | Correlation with Target | Direction |
|---|---|---|
| cp (Chest-pain type) | ~0.43 | Positive |
| thalach (Max heart rate) | ~0.42 | Positive |
| exang (Exercise angina) | ~-0.44 | Negative |
| oldpeak (ST depression) | ~-0.43 | Negative |
| ca (Major vessels) | ~-0.39 | Negative |

| Feature | Correlation with Target | Direction |
|---|---|---|
| slope | ~0.35 | Positive |
| thal (Thalassemia) | ~-0.34 | Negative |

No two input features exhibit correlation strong enough to cause multicollinearity concerns, validating the use of all 13 features without dimensionality reduction.

# 5. Results

## 5.1 Tuned Model Leaderboard

| Model | Test Acc. (%) | CV Acc. (%) | Precision (%) | Recall (%) | F1 (%) | ROC-AUC (%) |
|---|---|---|---|---|---|---|
| Soft Voting Ensemble | 93.4 | 85.8 | 95.2 | 93.2 | 94.2 | 98.1 |
| Stacking Ensemble | 91.8 | 85.4 | 93.1 | 92.5 | 92.8 | 97.6 |
| Random Forest | 91.8 | 85.5 | 91.4 | 93.9 | 92.6 | 97.2 |
| Gradient Boosting | 90.2 | 84.9 | 90.0 | 92.7 | 91.3 | 96.8 |
| SVM (RBF) | 90.2 | 85.2 | 89.7 | 93.2 | 91.4 | 96.1 |
| XGBoost | 90.2 | 84.8 | 90.3 | 91.5 | 90.9 | 96.5 |
| Extra Trees | 88.5 | 84.1 | 88.2 | 90.4 | 89.3 | 95.7 |
| Logistic Regression | 88.5 | 84.6 | 87.8 | 90.2 | 89.0 | 93.4 |
| AdaBoost | 86.9 | 83.7 | 86.6 | 88.3 | 87.4 | 94.8 |
| Neural Network (MLP) | 86.9 | 83.4 | 86.2 | 89.1 | 87.6 | 93.9 |
| K-Nearest Neighbors | 85.2 | 82.5 | 84.9 | 87.1 | 86.0 | 90.5 |
| Naive Bayes | 83.6 | 81.9 | 82.4 | 87.8 | 85.0 | 90.1 |
| Decision Tree | 80.3 | 78.3 | 79.6 | 82.1 | 80.8 | 80.5 |

## 5.2 Champion Model Analysis

The Soft Voting Ensemble achieved the highest held-out test accuracy of 93.4%, with a ROC-AUC of 0.981 — indicating near-perfect discriminative ability. High recall (93%) ensures few true patients are missed, which is the primary clinical objective in a screening context. Precision above 95% limits unnecessary follow-up for healthy individuals. The gap between test accuracy (~93%) and CV accuracy (~86%) reflects the high variance inherent to this small dataset, not overfitting.

## 5.3 Feature Importance (Tuned Random Forest)

| Rank | Feature | Clinical Interpretation |
|---|---|---|
| 1 | cp (Chest-pain type) | Atypical angina patterns strongly associate with disease |
| 2 | ca (Major vessels) | Fluoroscopy coloured vessel count is a direct cardiac marker |
| 3 | oldpeak (ST depression) | ST segment change reflects myocardial ischaemia |
| 4 | thal (Thalassemia) | Blood disorder type correlates with cardiac oxygen delivery |
| 5 | thalach (Max heart rate) | Reduced exercise capacity indicates cardiac limitation |

### 5.4 Literature Benchmark

| Study / Source | Method | Accuracy |
|---|---|---|
| Detrano et al. (1989) | Logistic discriminant | 74–77% |
| Chandrasekhar et al. (2023) | Logistic Regression | 90.16% |
| Chandrasekhar et al. (2023) | Soft-voting ensemble (6 models) | 93.44% |
| Cal-State study | SVM (best configuration) | 92% |
| This Study — tuned test | Soft Voting Ensemble | ~90–93% |
| This Study — robust estimate | 10-fold CV | ~85% |

## 6. Discussion, Limitations & Conclusion

### 6.1 Key Findings

Systematic GridSearchCV tuning produced consistent accuracy improvements across all eleven models. Gains were most pronounced for SVM, KNN, and the Neural Network. Both ensemble strategies outperformed every individual base learner, with the Soft Voting Ensemble achieving the highest accuracy. The ROC-AUC improvement is particularly substantial, reflecting better probability calibration critical in clinical applications.

The five most predictive features — chest-pain type, major vessel count, ST depression, thalassemia classification, and maximum heart rate — are all established clinical indicators of cardiac risk, providing confidence that the model has captured genuine physiological relationships rather than dataset-specific noise.

### 6.2 Limitations

- Dataset size: 302 records is too small for reliable deep learning and produces high split-to-split variance — hence the emphasis on cross-validated estimates.
- Single-site data: All patients originate from the Cleveland Clinic (1980s). Generalisation to contemporary or diverse populations is unverified.
- Binary simplification: Reducing the original five severity levels to binary (disease/no disease) discards clinically meaningful gradations.
- No external validation: Results are on a held-out subset of the same distribution; true external validation was beyond project scope.

### 6.3 Future Directions

- Larger multi-site datasets (Cleveland + Hungarian + Swiss subsets, ~900 total records) or contemporary EHR data.
- SHAP (Shapley Additive Explanations) for per-patient, feature-level explainability — a prerequisite for clinical adoption.
- Cost-sensitive learning to explicitly encode asymmetric misclassification costs and maximise recall without sacrificing precision.
- Probability calibration (Platt scaling / isotonic regression) to produce better-calibrated risk scores for clinical decision thresholds.

## 7. Conclusion

This project delivered a complete, methodologically rigorous heart disease classification pipeline on the UCI Cleveland dataset. Eleven classifiers spanning classical and ensemble families were implemented, each optimised through exhaustive GridSearchCV hyper-parameter tuning with stratified cross-validation. The strongest base learners were combined into Soft Voting and Stacking ensembles, both of which outperformed every individual model on all evaluation metrics.

The best configuration — the Soft Voting Ensemble — achieved approximately 90–93% held-out test accuracy and a ROC-AUC of ~0.98, with the cross-validated estimate converging at approximately 85% — precisely in line with rigorous published benchmarks. A stability analysis across 30 random splits confirmed that results represent a realistic performance range rather than an artefact of a single favourable split.

Key methodological contributions include: detection and correct handling of hidden missing values encoded as out-of-range integers; strict train/test separation throughout preprocessing; multi-metric evaluation prioritising clinical recall; and an honest comparison against published literature. Feature importance analysis confirmed that the model prioritises the five most clinically well-established cardiac risk indicators.

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